

Practices in visual computing 1

Lab 1: Python Basics

Subjects

1. Intro and Logistics
2. Basic Python
3. Intermediate Python
4. Basic Numpy

TA

- Send email to:

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- Office hours:

Wednesdays
1-2pm
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Ph.D. in CS

Assignments

1. Classical image processing
2. Image segmentation
3. Object detection
4. 3D computer vision

Project Proposal

Tips for the course

- Preferably, use VSCode (or other IDEs) for projects and Notebooks for snippets
- Use conda to manage environments and packages
- Avoid using ChatGPT for assignments. But definitely use it as a learning tool!
- You may be asked to write code snippets during assignment demos
- Avoid loops whenever possible
- Stick to Python best practices:
 - Proper namings, PEP8, Docstrings, etc
 - Try to seek optimal yet clean and easy-to-understand solutions

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Github Repo

<https://github.com/alimohammadiamirhossein/course-cmpt742-materials>

Question #1

String Reversal

- ❖ Write a Python function that takes a string as input and returns the string reversed.
- ❖ Follow-up: How can you do this without using slicing?

Question #2

Fibonacci Sequence

- ❖ Write a Python function that returns the n^{th} Fibonacci number.
- ❖ Follow-up: How can you optimize this function to avoid recalculating previously computed Fibonacci numbers?

Question #3

Palindrome Check

- ❖ Write a function that checks whether a given string is a palindrome (reads the same forwards and backwards).
- ❖ Follow-up: How would you modify this function to ignore spaces and capitalization?

Question #4

Sum of Two Numbers

- ❖ Given a list of numbers and a target number, write a function that finds two numbers in the list that add up to the target number. If no such pair exists, return None.
- ❖ Follow-up: What's the time complexity of your solution? Can you improve it?

Question #5

Prime Number Check

- ❖ Write a Python function to check whether a given number is prime.
- ❖ Follow-up: Can you optimize your solution to make it more efficient for large numbers?

Question #6

Calculating Weighted Sums

- ❖ Write a Python function to calculate the weighted sum of two lists (weights and values), using `random.uniform(0, 1)` for lists with 1 million elements.
- ❖ Follow-up: Redo the calculation using list comprehension and experiment with different random seeds to observe the effect on the results.

Question #7

Merge Sorted Lists

- ❖ Write a Python function that merges two sorted lists into one sorted list.
- ❖ Follow-up: What is the time complexity of this function?

Question #8

Anagram Detection

- ❖ Write a Python function that checks if two strings are anagrams (i.e., contain the same characters in a different order).
- ❖ Follow-up: How would you account for case sensitivity and spaces?

Question #9

Longest Substring Without Repeating Characters

- ❖ Write a Python function that finds the length of the longest substring in a given string that does not contain repeating characters.
- ❖ Follow-up: What is the time complexity of your solution? How can it be improved?

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Question #10

Dictionaries

- ❖ You are given a dictionary with student IDs as keys and their assignment grades (e.g., "a1": 95) as values. Write a function that calculates the average grade for each student and adds it to their dictionary as "avg".

Question #11

CSV and JSON

- ❖ For the previous question, construct the input dictionary from a .csv file.
 - Example format: std_id, a1, a2, a3, a4
- ❖ After calculating the averages, write them into a new .csv file.
 - Output format: std_id, a1, a2, a3, a4, avg

Question #12

Create a class `Student` that gets `std_id`, `name`, `age`, `gender`, and `major` as constructor.

It should also create a list of dictionaries, called `courses`.

A method `add_course` will receive a `course_name: grade` key-value pair and will append it to the `courses` list.

A method `get_gpa` will return the student's GPA by calculating the average for all courses.

A method `export_info` will create a `.txt` file containing human-readable information about the student.

© **Advanced tip:** Use `__str__` for custom string representation of the class.

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Question #13

Matrix Multiplication

Create the following two ndarrays

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{pmatrix}$$

And calculate the following:

$$AB \quad BA$$

Question #14

Slicing

First create a random 10x10 matrix, then using array slicing, change it to the following format:

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & x & x & x & x & x & x & 0 & 0 \\ 0 & 0 & x & x & x & x & x & x & 0 & 0 \\ 0 & 0 & x & x & x & x & x & x & 0 & 0 \\ 0 & 0 & x & x & x & x & x & x & 0 & 0 \\ 0 & 0 & x & x & x & x & x & x & 0 & 0 \\ 0 & 0 & x & x & x & x & x & x & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Question #15

Calculating Weighted Sums with Numpy

Redo the question six, this time using Numpy arrays. Use time module to compare the speedup between list and ndarrays.